# Meeting Assistant
# Roadmap & Future Work

Document 3 of 3 · Product roadmap, advanced capabilities, and research directions · Overview · Implementation · Changelog

This document consolidates everything that comes *after* the MVP: the sprint-by-sprint product roadmap, the three advanced differentiating capabilities (facilitator guide, document alignment, sentiment), the operational quality loop (MLOps), and a set of longer-horizon research ideas. It is the "future ideas" companion to Document 1 (vision) and Document 2 (the system as built today).

> **Status legend.** Sprint 1 (the MVP) is complete. Within the advanced capabilities, the **facilitator guide** is already implemented ( `GET /api/meetings/{id}/facilitation` ); document alignment and sentiment analysis are designed but not yet built.

> **Permanently out of scope:** live streaming speech-to-text during a meeting and professional speaker diarization. Single-shot Gemini transcription of uploaded/recorded audio covers the product need without a separate STT pipeline.

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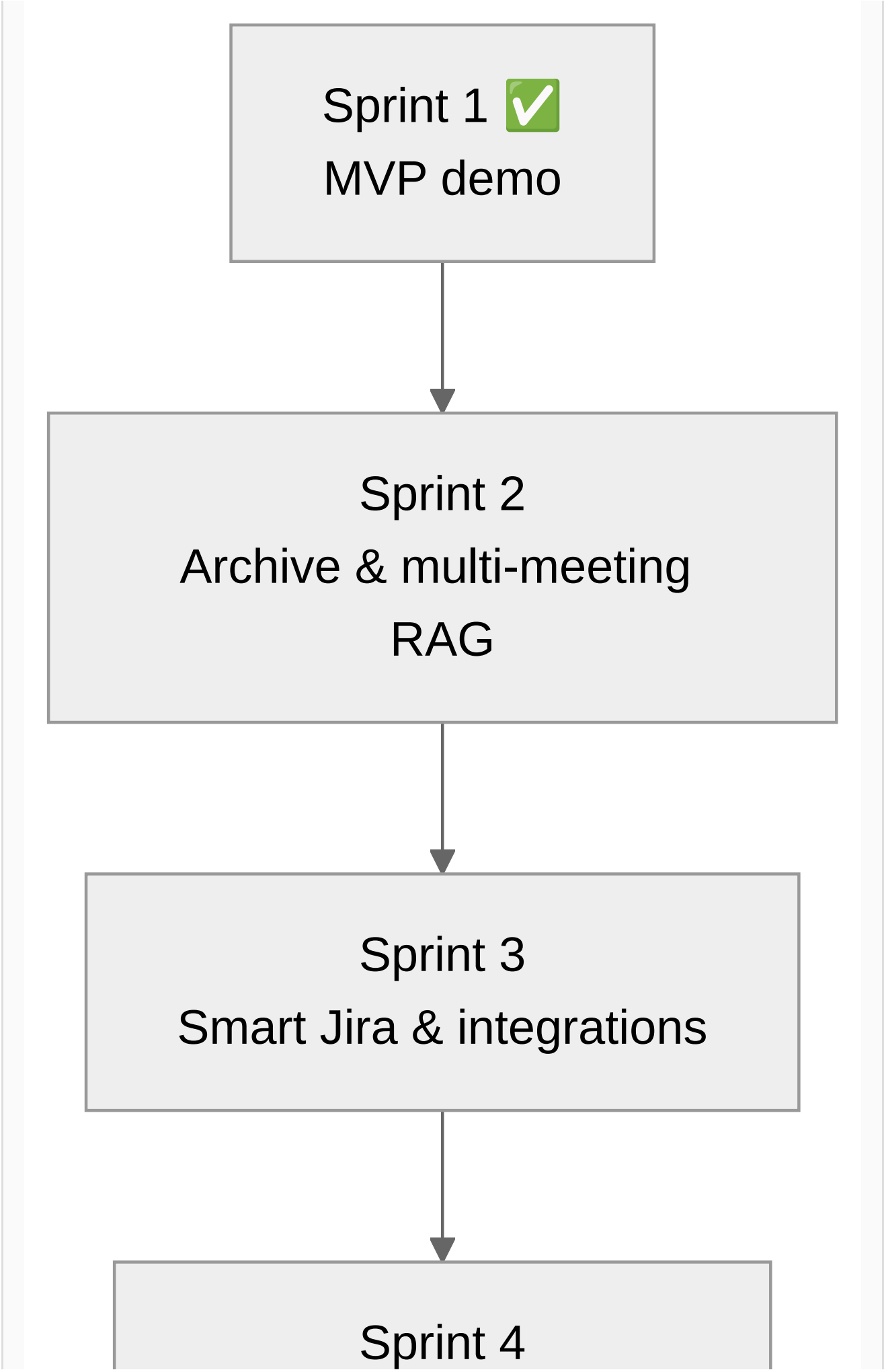
## 1. Roadmap at a Glance


Sprint 1 ✅
MVP demo
Sprint 2
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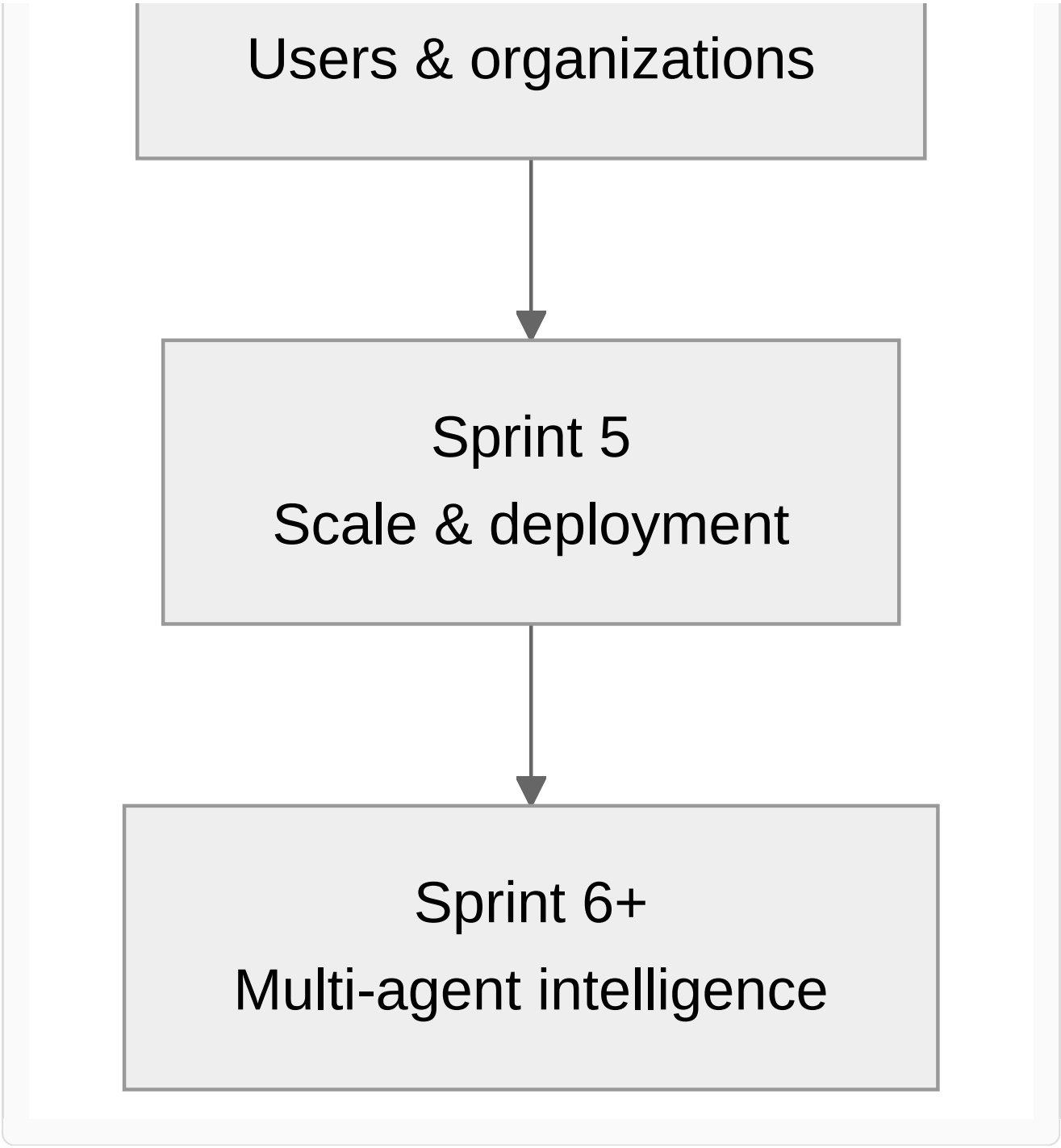



The arc of the roadmap is a deliberate progression: from a single meeting (MVP) to an organizational memory (Sprint 2), to real operational workflow (Sprint 3), to a true multi-tenant product (Sprints 4–5), and finally to differentiated multi-agent intelligence (Sprint 6+).

## 2. Sprint 2 — Organizational Archive & Search

**Goal:** move from "one meeting" to "an organizational memory of meetings."

| Capability | Description | Proposed implementation |
| --- | --- | --- |

| | | |
|---|---|---|
| Multi-meeting RAG | Ask across many meetings, filtered by date, project, or participant. | Chroma metadata `org_id`, `project`, `date`; filter at query time. |
| Archive page | List meetings with full-text search over title/summary. | SQLite FTS5 on `title`, `summary`. |
| Tags & projects | Categorize meetings (stand-up, planning, review). | `tags`, `project_key` columns in SQLite. |
| Summary export | Export to Markdown / PDF to share with managers. | Astro template or WeasyPrint from the same `MeetingAnalysis`. |
| Meeting comparison | "Last sprint's decisions vs. now" — an agent over multiple meeting ids. | `compare_agent` + top-k from each meeting. |

### Acceptance criteria

- Three meetings ingested; one cross-meeting question cited from ≥ 2 meetings.
- Archive search under 500 ms (up to ~100 meetings).
- One meeting's summary exported with no RTL rendering errors.

## 3. Sprint 3 — Jira & Operational Workflow

**Goal:** tasks actually reach the right people in Jira, not anonymous issues.

| Capability | Description | Technical |
|---|---|---|
| Assignee mapping | "Sara" in the transcript → a Jira `accountId`. | `speaker_jira_map` table; settings UI (already scaffolded). |
| `jira_agent` | Preview, de-duplicate, suggest epic/link. | Pydantic-AI tools: `search_existing_issues`, `preview_create`. |
| Jira OAuth | Replace the personal API token — team-friendly. | Atlassian OAuth 2.0; encrypted refresh token storage. |

| | | |
|---|---|---|
| Edit before send | Edit tasks in the UI, then bulk create. | Temporary state; POST only after confirmation. |
| Sync status | Show the issue key and Jira link on the meeting page. | `jira_keys_json` per task. |
| Issue templates | Project custom fields (story points, component). | Map from `JIRA_PROJECT_KEY` + a YAML config. |

## 4. Sprint 4 — Users, Workspaces & Calendar

**Goal:** real multi-user use and connection to everyday tools.

| Capability | Description | Technical |
|---|---|---|
| Authentication | Email/password login or organizational SSO. | JWT sessions, or OIDC (Keycloak / Google Workspace). |
| Workspaces | Each organization's data is isolated — meetings, speaker map, Jira. | `org_id` on every table; tenancy middleware. |
| Roles (RBAC) | admin · editor · viewer — who may create Jira issues. | `memberships` table. |
| Calendar (metadata) | Import title/time/attendees from Google Calendar — transcript still pasted/uploaded. | Read-only Calendar API; pre-fill the meeting form. |
| Summary email | Auto-send the summary + meeting link to attendees. | SMTP or Resend/Cloudflare Email; an RTL HTML template. |
| Slack/Teams alert | Short message: N new tasks + link. | Incoming webhook. |

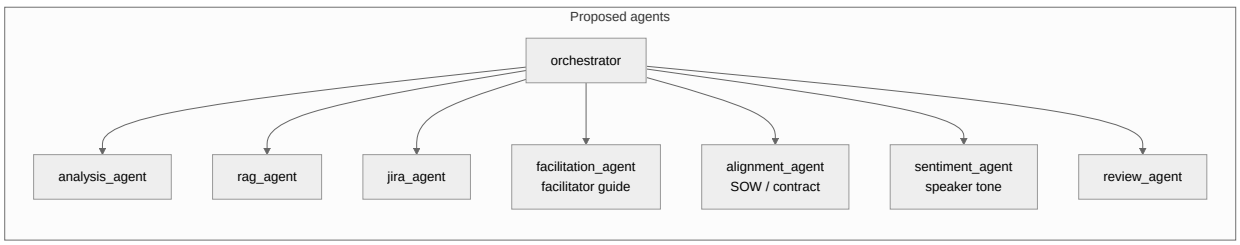
## 5. Sprint 5 — Scale, Quality & Deployment

**Goal:** ready for an organizational demo and production monitoring.

| Capability | Description | Technical |
|---|---|---|
| PostgreSQL | Replace SQLite for multiple workers and backups. | SQLAlchemy + Alembic migration from the current schema. |
| Vector at scale | Thousands of chunks without slowdown. | pgvector or Chroma Cloud; a retention policy. |
| Ingest queue | Analyze long meetings in the background. | Redis + worker (Celery / ARQ); UI status polling. |
| Observability | Track Gemini latency and Jira errors. | OpenTelemetry + structured logs; a dashboard. |
| Deployment | Docker Compose; optional Cloud Run / VPS. | frontend + API + Postgres in one compose file. |
| Eval & prompts | A golden-transcript set; quality regression on summaries. | pytest + an LLM-judge score or a human rubric. |
| Rate limiting | Prevent API abuse. | Per-org rate limit; embedding cache. |

## 6. Sprint 6+ — Intelligent Organizational Product

**Goal:** competitive differentiation — beyond "summary + Jira."

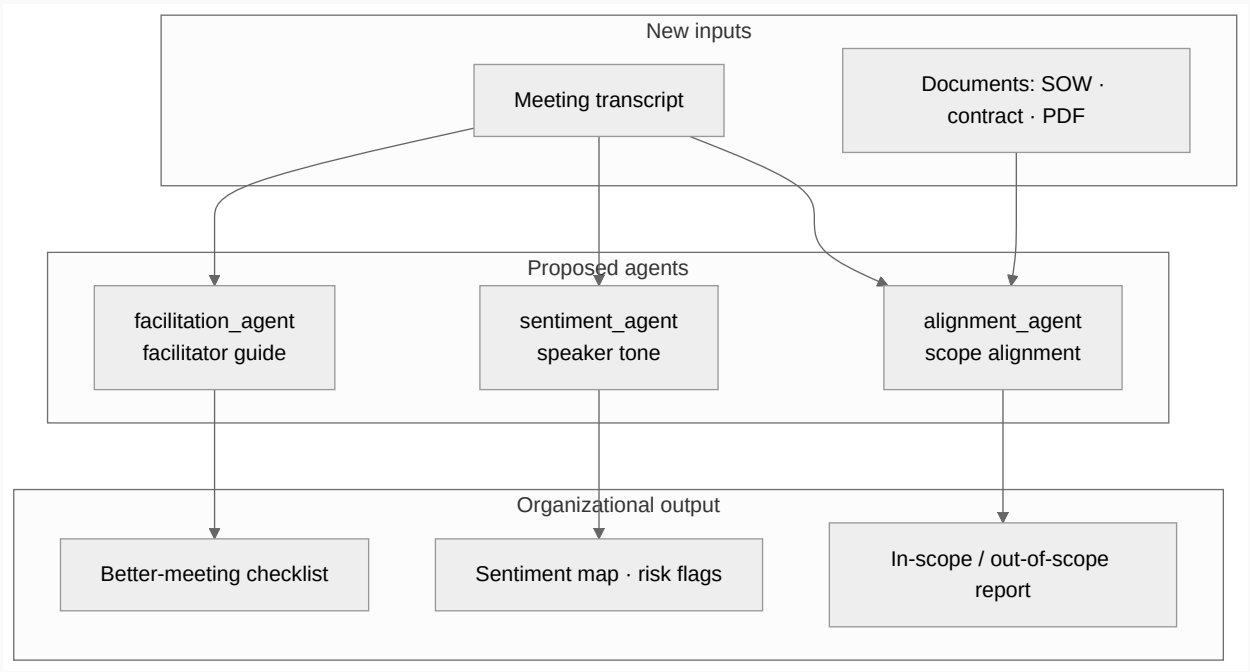



| Idea | Value |
|---|---|
| Decision tracking | Follow decisions across meetings — "was decision X carried out?" |

| | |
|---|---|
| Action-item dashboard | All open tasks from all meetings, filtered by assignee. |
| Meeting templates | Prompts specialized for stand-up / retro / planning. |
| Participation analysis | Each speaker's share of talk time; warn on one-sided meetings. |
| Decision conflict | An agent that compares a new meeting against past decisions. |
| Follow-up suggestion | Auto-draft the next agenda from open tasks. |
| Confluence / Notion | Publish minutes to the organization's wiki. |
| Analysis versioning | Human edits to a summary; re-run only part of the pipeline. |
| Privacy | PII masking · retention · GDPR export. |
| On-prem | In-network deployment; optional self-hosted model. |

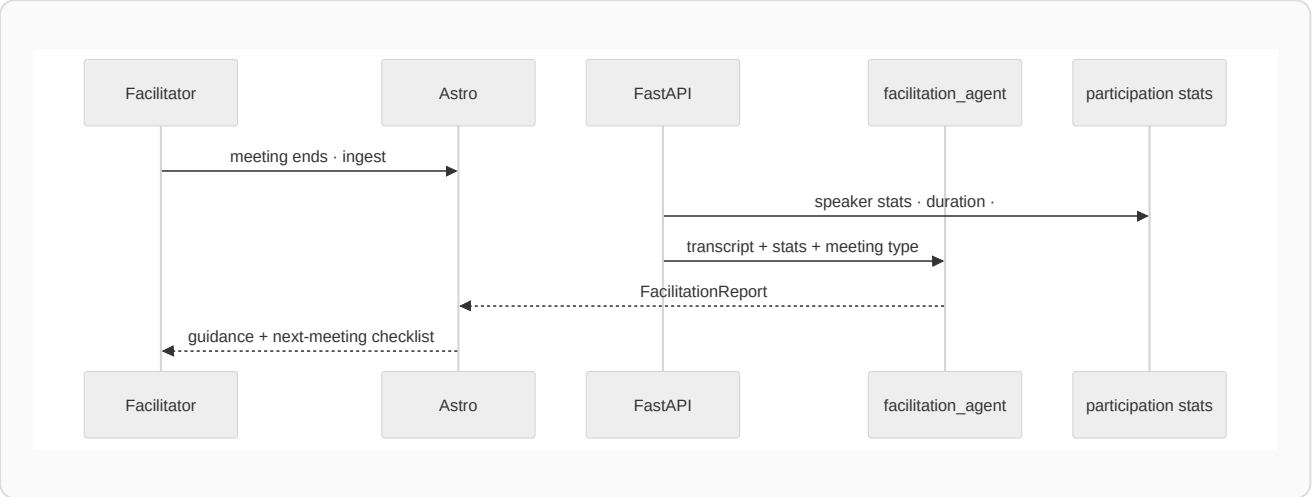
## 7. Advanced Capabilities (Differentiators)

Three capabilities turn the assistant from a note-taker into a meeting-intelligence product. Each adds a new agent and, where relevant, a new input.



## 7.1 Facilitator Guide *(implemented)*

After each meeting the system gives the **facilitator** concrete, coaching-style advice: what went well, where time was wasted, and how to run the next meeting shorter and more effectively.




### Manual checklist (usable today, no AI)

#### Before the meeting

- ☐ State a one-line goal and the expected output (decision / task / information).
- ☐ Send an agenda with a rough timebox per item.
- ☐ Invite only the necessary people; make each person's role clear.
- ☐ Make reference documents (SOW, contract) available in advance.

#### During the meeting

- ☐ State decisions explicitly: "we decided that…" with an owner.
- ☐ Send off-topic threads to a "parking lot."
- ☐ Give a 30-second progress recap every 15–20 minutes.

#### After the meeting

- ☐ Publish the summary + tasks within 24 hours.
- ☐ Schedule a follow-up only if there is an open task.

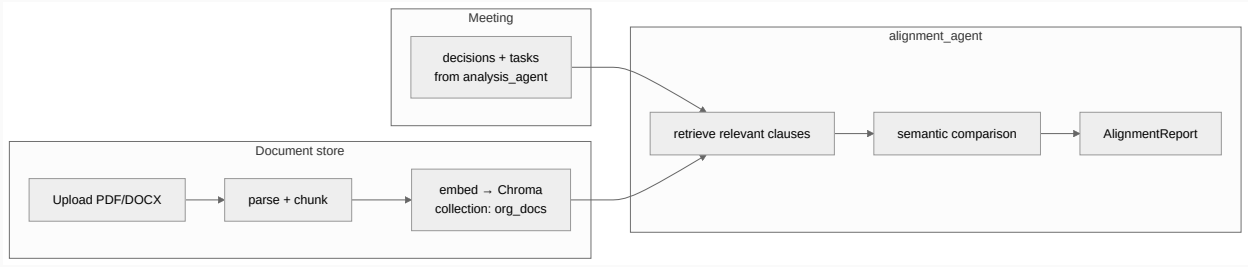
### FacilitationReport output

| Field | Description |
|---|---|
| `what_went_well` | 2–3 positives (e.g. clear decisions, balanced participation). |
| `improvements` | Specific suggestions (e.g. "section X took 40% of the time with no decision"). |
| `next_meeting_agenda` | A draft agenda for the next meeting, from open tasks. |
| `timebox_suggestion` | A shorter duration or a different format (e.g. async update). |
| `coaching_summary` | A short, encouraging summary for the facilitator. |
| `facilitator_score` | An optional 1–5 score against a rubric. |

**API:** `GET /api/meetings/{id}/facilitation` — implemented, with a "Facilitator Guide" tab in the UI.

### 7.2 SOW / Contract Alignment *(planned)*

An organization can upload a **Statement of Work**, contract, or RFP. After each meeting, `alignment_agent` compares the meeting's decisions and commitments against the document's clauses and labels each as **in-scope**, **out-of-scope**, or **needs legal/manager approval**.




#### Proposed data model

| Entity | Fields |
|---|---|
| `OrgDocument` | `id`, `title`, `type` (sow\|contract\|rfp), `project_key`, `uploaded_at` |

| | |
|---|---|
| `AlignmentFinding` | `meeting_decision`, `doc_excerpt`, `status` (aligned\|out_of_scope\|unclear), `confidence`, `doc_id` |
| `AlignmentReport` | A list of findings + a summary for the project manager. |

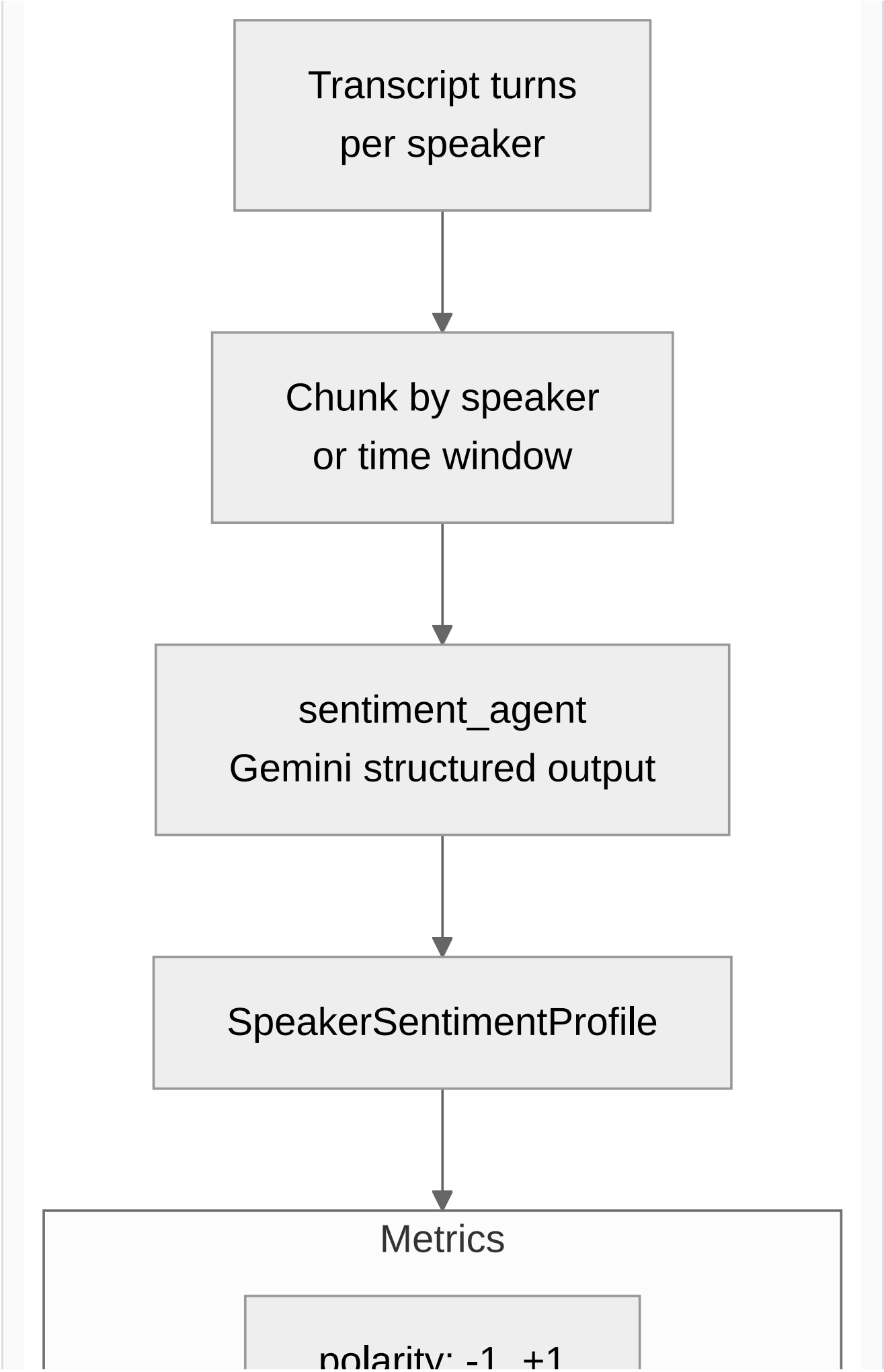
**Proposed API**

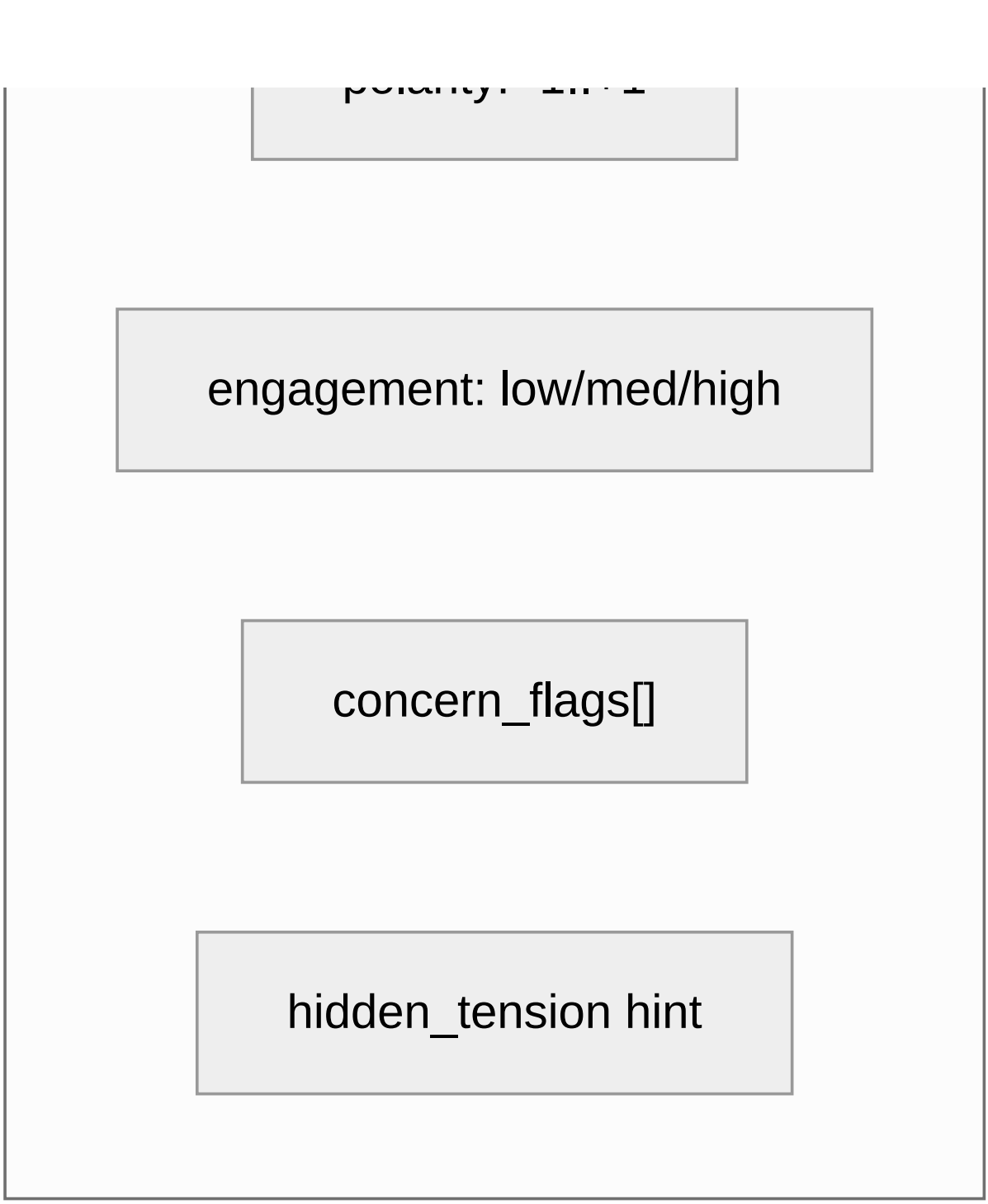
| Method | Path | Purpose |
|---|---|---|
| POST | `/api/documents` | Upload an SOW/contract · index in Chroma. |
| GET | `/api/documents` | List a project's documents. |
| POST | `/api/meetings/{id}/alignment` | Run `alignment_agent` · comparison report. |

> **Product caveat:** AI output does not replace legal counsel — for binding contracts, human approval is mandatory. The UI must show a clear disclaimer.

### 7.3 Sentiment Analysis *(planned)*

Over the **transcript text** (not raw audio), for each speaker and each phase of the meeting: overall tone (positive / neutral / negative), **incongruity** (e.g. positive words hiding a concern), and the trend of sentiment over the meeting.


Transcript turns
per speaker
Chunk by speaker
or time window
sentiment_agent
Gemini structured output
SpeakerSentimentProfile
Metrics
polarity: -1 +1




**SpeakerSentimentProfile output**

| Field | Meaning |
| --- | --- |
| `speaker` | Speaker name from the transcript. |
| `overall_tone` | positive \| neutral \| negative \| mixed. |
| `polarity_score` | A number from −1 to +1. |
| `concern_phrases` | Short quotes that hint at concern or disagreement. |

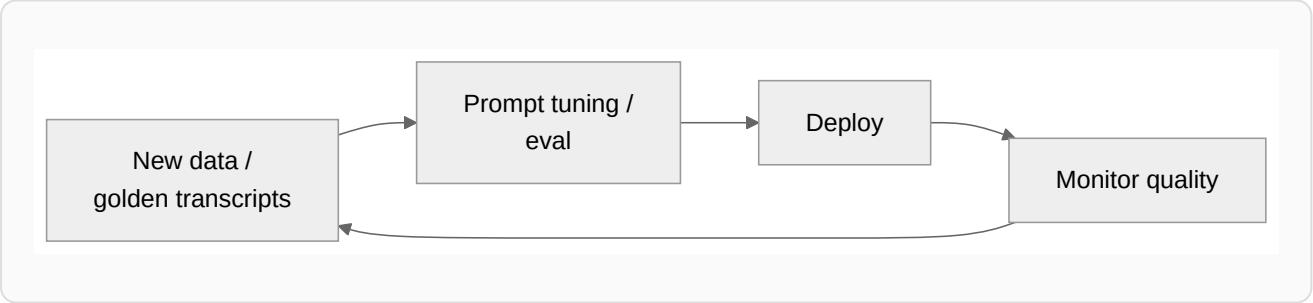
| | |
|---|---|
| `hidden_dissatisfaction` | bool + explanation (e.g. "verbal agreement but stress on risk"). |
| `timeline` | Tone change across the start / middle / end of the meeting. |

**Proposed API:** `GET /api/meetings/{id}/sentiment` — a new tab beside summary/tasks/RAG.

> **Ethics & privacy:** sentiment analysis for team management is sensitive. Only an admin/facilitator should see it; it must be organizational opt-in, with no negative labels in public reports. Framing in the UI: "a signal for a follow-up conversation," never "a personality judgment."

## 8. MLOps & the Quality Loop

The MVP does not retrain a model — it consumes a hosted LLM. The operational concern, therefore, is not model training but **quality assurance of prompts and outputs over time**, which is exactly where an AI product tends to regress silently.




- **Golden set:** a curated set of transcripts with reference summaries and expected tasks.
- **Automated eval:** score each new prompt/model version against the golden set with an LLM-judge and/or a human rubric, run as part of CI.
- **Prompt versioning:** treat prompts as versioned artifacts so a regression can be traced to a specific change.
- **Drift monitoring:** watch summary quality, RAG citation accuracy, and Jira conversion over time, and alert on degradation.

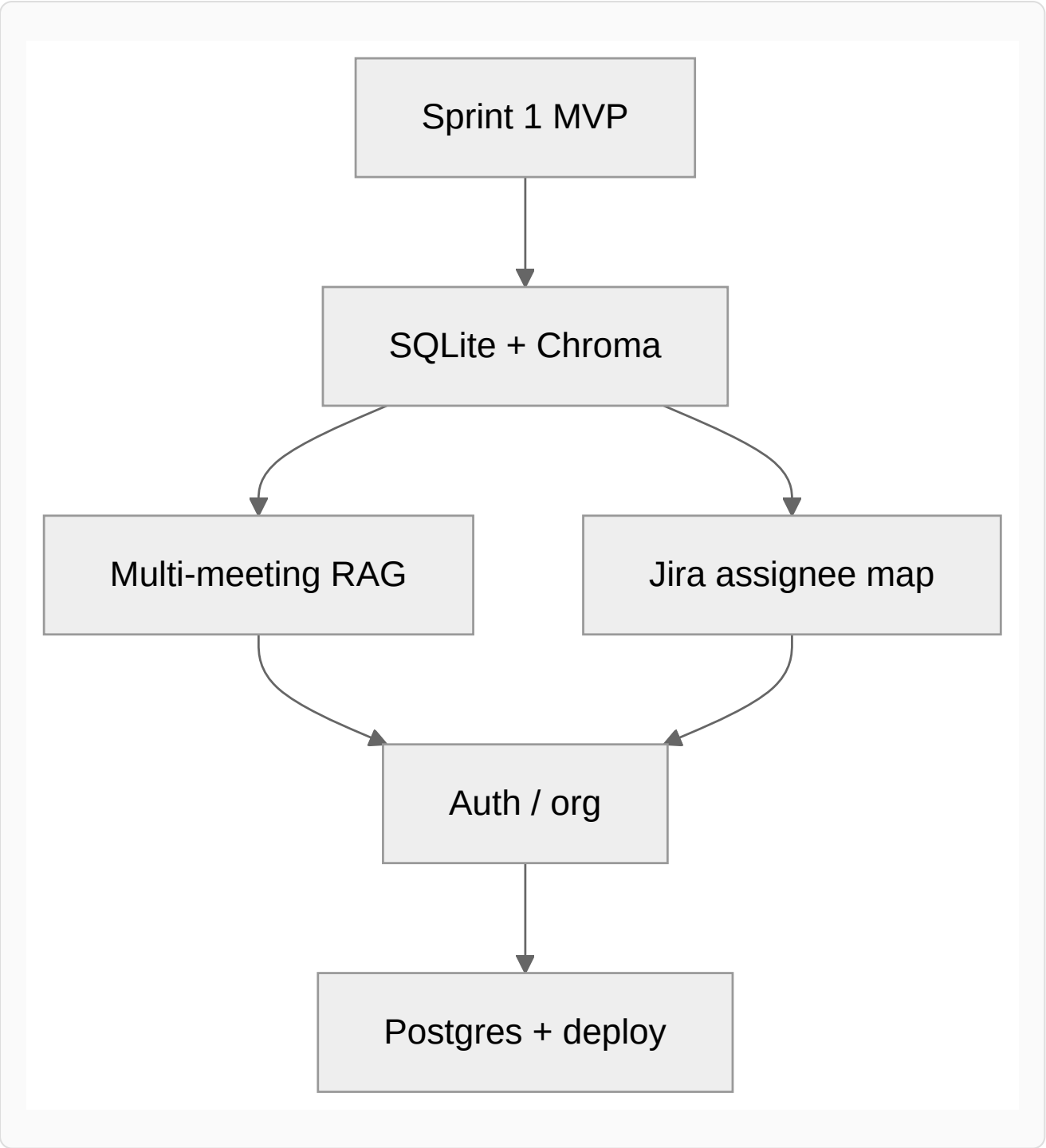
This is scheduled with Sprint 5 (Scale, Quality & Deployment) and is the honest counterpart to the AIOps/MLOps theory: the lifecycle the product would adopt as it matures, scoped

realistically rather than over-promised in the MVP.

## 9. Prioritization

| Priority | Item | Rationale |
|---|---|---|
| P0 | Multi-meeting RAG + FTS archive | Highest value over the MVP. |
| P0 | Jira assignee mapping | A real pain point for operational teams. |
| P1 | Auth + workspace | Prerequisite for an organizational customer. |
| P1 | Edit-before-create for Jira | User trust in AI output. |
| P1 | Facilitator guide (done) + deeper coaching | Immediate differentiation, light infrastructure. |
| P1 | SOW/contract alignment | Value for PMs and fixed-scope contracts. |
| P2 | PostgreSQL + queue | When traffic exceeds demo scale. |
| P2 | Calendar metadata + summary email | Reduces data-entry friction. |
| P2 | Sentiment analysis | Only after RBAC and ethical opt-in. |
| P3 | Multi-agent · Confluence | Long-term differentiation. |

## 10. Technical Dependencies




The dependency graph explains the sprint ordering: the storage foundation enables both multi-meeting RAG and assignee mapping; both of those motivate multi-tenancy (auth/org); and a real tenant load is what finally justifies PostgreSQL and a production deployment.

## 11. Longer-Horizon Research Ideas

Beyond the planned product work, several directions would make strong follow-on research or thesis extensions:

- **Cross-meeting decision graphs.** Model decisions as nodes and detect contradictions, reversals, and unfulfilled commitments across an organization's meeting history.
- **Multi-agent debate for summaries.** Use a critic/reviewer agent to challenge the first-pass summary, measuring whether adversarial review improves factual faithfulness.
- **Semantic, speaker-aware chunking.** Compare turn-based chunking against semantic and speaker-aware chunking on retrieval accuracy for Persian transcripts.
- **Faithfulness evaluation for Persian RAG.** Build an evaluation harness for citation accuracy and hallucination rate specific to low-resource-language meeting data.
- **Privacy-preserving analysis.** On-device or PII-masked pipelines so sensitive meetings can be analyzed without leaving the organization's network.
- **Human-in-the-loop quality.** Capture human edits to summaries and tasks as a feedback signal for prompt improvement and per-organization adaptation.

Related: Project Overview & Vision · Technical Implementation.